

TED UNIVERSITY

Department of Computer Engineering

CMPE 491 – Senior Design Project I

PROJECT PROPOSAL

Improving Object Detection Performance in SAR Imagery Using Super-Resolution Techniques

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Project Website
https://ilkeozcendek.github.io/keskin/

Related Program
SAYZEK-ATP 2026–2027 · Theme 3, Topic 28
[under evaluation]

1. Project Description

1.1 Problem Definition and Objectives

Synthetic Aperture Radar (SAR) imaging allows for day-and-night as well as all-weather observation, which is critical for disaster response and critical infrastructure monitoring. Object detection in SAR images is, however, fundamentally limited by imaging-system constraints: compared to optical images, SAR data is affected by speckle noise, low spatial resolution, and complex backscattering, which obscure fine spatial details and blur the boundaries of small or closely packed targets.

These resolution constraints severely limit the performance of modern deep learning object detection architectures, causing high false alarm rates and the missed detection of crucial targets such as vehicles and infrastructure facilities. However, purchasing high-resolution SAR data directly is often cost-prohibitive or constrained by satellite revisit times. Deep learning-based super-resolution offers a promising computational approach to enhance image clarity and retrieve high-frequency spatial details from readily available open-source datasets.

Research Question

How does deep learning based super resolution pre-processing improve object detection performance on SAR imagery?

Objectives

O1

Construct a paired lower to higher resolution SAR dataset using open-access data such as Copernicus Sentinel-1 (C-band) and Umbra (sub-meter X-band), pre-processed with tools like ESA SNAP and QGIS.

O2

Implementing and comparing at least two deep learning super-resolution architectures (e.g., EDSR and SRGAN) tailored for SAR and characteristics.

O3

Integrate super-resolved images into object detection models (e.g., YOLO, CNN) to evaluate low resolution inputs.

1.2 Proposed Approach

1.2.1 Dataset and Image Pairs

This project will use the open-access dataset prepared for ship detection. It contains SAR images and from satellites, which covers the Sentinel data target. Among the targets mentioned in Section 1.1, ships were chosen because open annotated SAR data is available for them and they appear small in images showing wide areas. The original images will serve as high-resolution (HR) references and will be blurred and downsampled by factors of 2 and 4 to create low-resolution (LR) images. The boxes showing the positions of the ships will also be adjusted to the new image sizes. If time permits pre-processing with ESA SNAP and QGIS, will be used for a qualitative test on real satellite images.

1.2.2 Super-Resolution Models

This project will compare two candidate models for increasing image resolution, corresponding to SR model A and SR model B in Table 1: EDSR, a CNN model that aims to reconstruct the image as close to the original as possible, and SRGAN, a GAN model that creates sharper images but can also create details that are not actually present. The final choice will be confirmed during model selection. Both models will be trained with SAR images, and their results will be compared with bicubic interpolation, which requires no training. The work of Addabbo et al. [2] shows that CNN models can reduce the loss of resolution in SAR images. In this project, we will investigate whether either model, especially the sharper GAN output, improves detection.

1.2.3 Object Detection Phase

The YOLOv8 model, which can be trained quickly, will be used to detect . Following the evaluation approach in Section 1.3, the model will be trained and tested separately for each input condition in Table 1, using the same settings and data splits. In addition, a model trained only on high-resolution images will also be tested on the other input conditions. The metrics used for this comparison are described in Section 1.3.

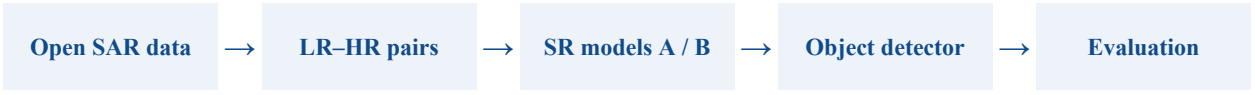


Figure 1. *Planned processing pipeline.*

1.3 Evaluation and Expected Outcomes

For the evaluation we will follow a controlled evaluation approach inspired by Shermeyer and Van Etten [3]. For each input condition in Table 1 we will train the same detector architecture while keeping the training settings keeping train and test splits unchanged. This will allow us to compare the results in a fair way. Image quality will be measured using PSNR and SSIM against corresponding high resolution reference images. We will evaluate the detection results using precision, recall and mean average precision (mAP) with (0.5) IoU threshold .

Better image quality does not always lead to better detection results. We will therefore present both sets of metrics together and look at how detection performance changes depending on object size.

Table 1. *Planned comparison conditions*

Input Condition	Role in the Comparison
Low-resolution input	Baseline without super resolution
Bicubic upsampling	Non learning upsampling baseline
SR model A	First learned super-resolution approach
SR model B	Second learned super resolution approach
High-resolution reference	Reference detection comparison in case if available

The architectures for SR models A and B will be finalised during model selection

Expected Outcomes

D1 Trained super resolution models for the selected SAR dataset, based on two architectures to be finalised.

D2 A complete and reproducible workflow that includes data preparation, super resolution, object detection and evaluation.

D3 A comparison to determine whether super-resolution improves detection performance compared to the bicubic baseline and under what conditions these improvements occur.

2. Initial Literature Review

2.1 Object Detection in SAR Imagery

[1] Li et al. · IEEE Transactions on Geoscience and Remote Sensing, 2017

> ***Summary:*** *The authors introduced OpenSARShip, a dedicated benchmark dataset constructed from Sentinel-1 C-band Synthetic Aperture Radar (SAR) imagery to evaluate ship detection and recognition algorithms across varied sea states and incidence angles. Using standard convolutional neural network (CNN) architectures alongside traditional algorithms, the study demonstrated that spatial resolution limitations directly contribute to high false alarm rates and poor target boundary delineation in complex backgrounds. This work establishes a baseline for open-source SAR target detection and directly justifies super-resolution pre-processing to reconstruct missing high-frequency spatial features prior to object detection. However, the study relies exclusively on Sentinel-1 C-band data without integrating higher-resolution X-band imagery.*

2.2 Super-Resolution for SAR Images

[2] Addabbo et al. · IEEE Access, 2023

Addabbo et al. developed a CNN model called DC2SCN to improve the resolution of SAR images. The model attempts to preserve both amplitude and phase information by processing the real and imaginary portions of SAR data separately. Researchers obtained low and high-resolution images by processing COSMO-SkyMed StripMap data at different bandwidths. With this method, approximately 20,000 image segments with a size of 512 × 512 pixels were obtained. In tests performed on real data, it was observed that the model increased the PSNR value by approximately 14 dB compared to the input images and brought the resolution of point targets to a level close to high-resolution images. This study shows that deep learning models

can reduce the loss of resolution in SAR images. In addition, the image generation method used can also be evaluated in the KESKİN project. However, the research only examined image quality, did not compare the model with other super-resolution methods, and did not test the contribution of the improved images to object detection. The KESKİN project investigates whether increasing image resolution improves object detection success.

2.3 Effect of Super-Resolution on Detection

[3] *J. Shermeyer and A. Van Etten · CVPR Workshops (EarthVision), 2019*

Shermeyer and Van Etten [3] looked at whether super resolution could improve object detection in optical satellite images. They used the same detectors for the original images and those enhanced with VDSR or a random forest model. The scaling factors ranged from 2× to 8× and the image resolutions varied between 30 cm and 4.8 m. When they increased the resolution from 30 cm to 15 cm, mAP improved by 13–36%. However the improvement was much smaller for images with lower resolutions. We can use a similar approach to evaluate our results but we also need to consider that SAR images contain speckle noise so the same improvements may not be seen in our study.

2.4 Research Gap

Super resolution has been shown to improve object detection in optical satellite images but we do not yet know if it will have the same effect on the SAR images used in our project. We will compare image quality and detection results across the different input conditions paying particular attention to small objects. Our aim is to see whether the improvements in image resolution also lead to better detection results.

References

[1] [1] R. Li, X. Zhang, and A. Author, "OpenSARShip: A dataset dedicated to Sentinel-1 ship interpretation," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 10, no. 11, pp. 5028–5039, 2017, doi: 10.1109/JSTARS.2017.2755838.

[2] P. Addabbo, M. L. Bernardi, F. Biondi, M. Cimitile, C. Clemente, N. Fiscante, G. Giunta, D. Orlando, and L. Yan, "Super-resolution of synthetic aperture radar complex data by deep-learning," *IEEE Access*, vol. 11, pp. 23647–23658, 2023, doi: 10.1109/ACCESS.2023.3251565.

[3] *J. Shermeyer and A. Van Etten, "The effects of super-resolution on object detection performance in satellite imagery," in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW), 2019. [Online]. Available: https://arxiv.org/abs/1812.04098*